

TSB_20260305

Date: 5 March 2026

Applies to: All QT/EPEC switchboards utilizing Siemens Molded Case Circuit Breakers

Subject: Interpretation of Contact Resistance Measurements for Low-Voltage Circuit Breakers

During commissioning and maintenance of low-voltage switchboards, some third-party testing organizations perform micro-ohm or millivolt-drop measurements across the main current path of circuit breakers. These tests are sometimes interpreted as an indicator of breaker condition or suitability for service. The information below, and in the following pages, clarifies how such measurements should be understood when evaluating Siemens molded case circuit breakers.

Manufacturers of low-voltage circuit breakers design their products to meet established, recognized standards, including UL 489 and related industry qualification procedures. These standards evaluate a breaker's ability to carry current, interrupt faults, and operate reliably under controlled laboratory conditions. Verification of breaker performance in the field is therefore intended to follow inspection and testing procedures developed specifically for molded case breakers, such as those outlined in NEMA AB-4.

Resistance measurements taken across a breaker's main current path can vary significantly depending on test configuration and environmental conditions. The measured value reflects not only the internal contact interface, but also multiple elements of the installed current path including terminals, conductor interfaces, test lead placement, and the mechanical pressure of the measurement probes. Additional factors such as oxidation, contaminants, temperature, previous switching operations, and the number of load interruptions can also influence the result. Because of these variables, measurements obtained during field testing may differ from phase to phase or test to test even when the breaker is functioning normally.

For this reason, static micro-ohm or millivolt-drop measurements are not considered a reliable method of determining the operational condition of low-voltage circuit breakers. Circuit breaker manufacturers do not publish acceptable resistance tolerances for these measurements, and they are not used as a basis for determining product conformity or warranty status. Resistance readings alone therefore should not be interpreted as evidence of a defective or unsuitable breaker.

Evaluation of breaker performance in the field should instead rely on the procedures described in NEMA AB-4 and related industry practices. These procedures focus on functional testing of protective operation and mechanical performance, including verification of trip characteristics, inspection of mechanical components, and confirmation of proper installation and connections. When these recommended methods are followed, they provide a meaningful assessment of breaker condition and serviceability.

In summary, while resistance measurements may occasionally be used as a general diagnostic reference during commissioning, they should not be considered a definitive indicator of breaker health. Variations in measured values are common and expected, and any evaluation of circuit breaker performance should be based on recognized industry testing procedures rather than contact resistance readings alone.

Attachments:

Siemens CBFL-CRPST-0619, *"Static Micro-ohm Resistance Readings for Circuit Breaker Main Contacts"*

Siemens White Paper, *"Field Testing of Siemens Circuit Breakers"*

Static Micro-ohm Resistance Readings for Circuit Breaker Main Contacts

There is a common misconception regarding resistance across the main contacts in the primary current path of power circuit breakers and molded case circuit breakers. The belief is that this resistance is an indicator of the general health of the circuit breaker, specifically its ability to reliably carry current without overheating, or avoid nuisance tripping. While in some cases, high contact resistance may be a symptom of a circuit breaker that has worn or damaged main contacts, the resistance value usually measured by many third-party testing companies is irrelevant, and often misleading in new or nearly new circuit breakers.

By design, Siemens low voltage power and molded case circuit breakers have internal main contacts that are self-cleaning when the circuit breaker operates under load. Particularly for new circuit breakers, but applicable for any circuit breaker that has been left open for a period of time, any contaminants are removed during loaded operations. In order to maintain this functionality in power and molded case circuit breakers it is important that the users of such circuit breakers not attempt to adjust, clean, dress, or polish the main contacts.

Siemens Industry (as well as other manufacturers of low-voltage circuit breakers) does not recommend performing milli-ohm or milli-volt drop tests across circuit breaker contacts because it has been shown that such tests are not a reliable or accurate method of determining the suitability of a given circuit breaker for its intended application.

The resistance of the main current path (including the contacts) in a circuit breaker is variable. Contact resistance, and the measurement of contact resistance is affected by:

- Climate
- Number and magnitude of past interruptions
- Foreign material between the contacts
- Oxidation of contact surfaces
- Assembly tolerances
- Where and how the readings are taken

Static micro-ohm or milli-volt drop readings are highly sensitive to all the previously mentioned criteria — a known and acceptable condition to Siemens and other manufacturers of power circuit breakers.

Contact resistance in the order of magnitude that is measured in the field (20 — 200 micro-ohms), aside from being variable, is negligible both in theory and in practice, as are the phase-to-phase differences frequently measured. The NETA document "Acceptance Testing Specifications" makes an arbitrary statement as to the allowable tolerance of this measurement when no tolerances are stated by the manufacturer. As applied to Siemens circuit breakers, this arbitrary pass/fail criterion has no basis in the practical operational environment of the circuit breaker.

For the reasons above, Siemens Industry neither publishes any tolerances for, nor recommends the testing of static micro-ohm or milli-volt drop across the main contacts of any Siemens low voltage power or molded case circuit breakers.

Siemens WL circuit breakers are qualified to UL 1066 and UL 489, as well as conforming to relevant recognized industry performance standards such as ANSI/IEEE C37.13, C37.16, C37.17, C37.20.1, and IEC 60947-2. The WL circuit breakers also meet all recognized industry testing and qualification requirements, such as ANSI C37.50, UL 489, and IEC 60947-2. Siemens molded case circuit breakers, including the 3VA, VL, Sentron, xGB/xGG and BQD/CQD families, are qualified and listed to UL489.

Siemens Industry does not accept the NETA "Acceptance Testing Specifications" as an accurate means of validating the quality or operational performance of its low voltage power or molded case circuit breakers. Any test report or claim stating that a Siemens circuit breaker is not suitable for its intended application or service on the basis of contact resistance or milli-volt drop, will not be considered acceptable evidence of nonconformity for warranty action by Siemens. Any other conformity issues or conditions will be considered under the Siemens Industry Standard Terms and Conditions of Sale.

April 7, 2011

To: General Distribution

Re: Field Testing of Siemens Circuit Breakers

Siemens, along with most other major manufacturers of circuit breakers, is a member of NEMA (the National Electrical Manufacturers Association). Like other members of NEMA, Siemens requires the use of the NEMA AB 3 and NEMA AB 4 standards in the evaluation of circuit breakers including the use of time current trip curves and field testing of circuit breakers Listed by UL.

The latest edition of AB 4 was developed and written by ABB Control Inc., Eaton Corporation, General Electric, Moeller Electric Corporation, Square D Company, Thomas & Betts Corporation, and Siemens Energy & Automation. The standard was then voted on and approved by the membership of NEMA which includes members from a wide cross-section of the electrical industry.

AB 4 is recognized and referenced as the standard for field testing by ANSI and IEEE. Specifically IEEE 1015-2006 "IEEE Recommended Practice for Applying Low Voltage Circuit Breakers in Industrial and Commercial Power Systems". Paragraph 3.34, states "NEMA AB 4 should be referenced for MCCB and ICCB field inspection and maintenance." ANSI lists this standard under the same number.

NEMA AB 3 states that time current trip curves are for application and coordination purposes only. Many of Siemens time current curves also contain this statement, and this statement applies to all Siemens trip curves even if not specifically shown. This statement deliberately excludes the use of time current curves for field testing. However, AB 4 does reference specific information from the breaker time current curve (TCC), along with requirements for the interpretation and use of the information which must be followed.

Time current curves developed according to AB 3 are based on 40 C ambient cold start, with current in all three poles, connected with 4 feet of rated wire per terminal, open air, under lab controlled conditions. These conditions are, for all practical purposes, impossible to duplicate in the field. Also there are variations in field testing configurations, i.e. cable length and size, test current in only one pole, pulse current source vs. run-up continuous current source, field testing equipment calibration, and field conditions in general. The tolerances established in AB 4 allow for these field variations and NEMA member circuit breaker manufacturers require the test procedures be followed for field testing. Siemens will not authorize the warranty return of a circuit breaker, based on field test alone, that is not tested per NEMA AB 4.

Table 3 below shows the proper current and trip times for the thermal tripping characteristics of the circuit breaker and must be used instead of the TCC.

Table 3		
VALUES FOR INVERSE TIME TRIP TEST		
(At 300% of Rated Continuous Current of Circuit Breaker)		
Rated Current Amperes	Maximum Trip Time in Seconds*	
	≤250V	251–600V
0–30	50	70
31–50	80	100
51–100	140	160
101–150	200	250
151–225	230	275
226–400	300	350
401–600	—	450
601–800	—	500
801–1000	—	600
1001–1200	—	700
1201–1600	—	775
1601–2000	—	800
2001–2500	—	850
2501–5000	—	900
6000	—	1000

* For integrally-fused circuit breakers, trip times may be substantially longer if tested with the fuses replaced by solid links (shorting bars).

AB 4 Section 6.6, Instantaneous Overcurrent Trip Test, details methods for these tests and Table 4, shown below, gives the required trip current tolerances.

Table 4			
INSTANTANEOUS TRIP TOLERANCES FOR FIELD TESTING OF CIRCUIT BREAKERS			
Breaker Type	Tolerances of Settings	Tolerances of Manufacturers' Published Trip Range	
		High Side	Low Side
Electronic Trip Units ⁽¹⁾	+30%	—	—
	-30%	—	—
Adjustable ⁽¹⁾	+40%	—	—
	-30%	—	—
Non-adjustable ⁽²⁾	—	+25%	-25%

⁽¹⁾ Tolerances are based on variations from the nominal settings.

⁽²⁾ Tolerances are based on variations from the manufacturer's published trip band (i.e., -25% below the low side of the band; +25% above the high side of the band).

Another potential issue is the testing of circuit breakers while installed in the end use equipment. The testing of the circuit breaker outside of the enclosure, i.e. 'on the bench', minimizes the potential variation of current paths and other test variables and AB 4 recommends that the breaker be removed from equipment for test.

However, the 2003 version of AB 4 does make a provision for installed breaker testing under certain conditions where "the circuit breaker can be safely isolated as installed . . ."

Should such a test be desired, the test operator must be responsible for the test setup. Among the possible variables that must be controlled or accounted for, the most important must be the current paths, including source current connections, shorting connections, and proper electrical isolation from other equipment. The test operator must also become thoroughly familiar with the end use equipment configuration and follow proper safety procedures. The field test tolerances shown in Table 3 and Table 4 should allow for potential variances introduced by installed testing.

However, due to wide variation of end use equipment, the increased complexity of testing of an installed breaker may make it difficult to achieve acceptable results. In this case Siemens requires that the breaker be removed from the equipment and the test repeated.

Finally, circuit breaker contact resistance testing is sometimes performed in the field. Siemens, in agreement with NEMA, does not recognize circuit breaker contact resistance alone as a reliable indicator of breaker performance and will not authorize a warranty return based solely on such a test. Any testing must be performed per AB 4 and must also be used in conjunction with other tests to determine if there is a potential breaker issue.

Please contact me if further information on this test is desired or if there are any other questions concerning field testing of Siemens circuit breakers.

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